
CondVQA: A Benchmark for Conditional Visual Question Answering in Video Understanding

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Abstract

We propose a benchmark for answering condition-based questions in a Visual Question Answering model. CondVQA is a multi-step model which is able to take a conditional question as an input and split it into condition and question. CondVQA is able to classify questions as SIMPLE (no condition), or CONDITIONAL (contains condition), through an LLM backbone. With few-shot learning, the LLM backbone learns the semantics of a conditional question, and is accurately able to pass both a condition and a SIMPLE question into the VLM. With such condition, a subset of frames can be isolated which in turn are handled by the subsequent SIMPLE question. Therefore, with a specified attribution probability method, it is able to give a leading answer based on the frames. Unlike other models, CondVQA focuses on splitting the input question first, instead of focusing on all frames of a video input. Our benchmark will be tested both on three Conditional Datasets created, and three General questions dataset. Among the datasets, there is a conditional benchmark dataset and 2 general benchmark datasets. Our code is available at <https://github.com/rdgbrandon/CondVQA>.

1 Introduction

Video-Question-Answering models have become a staple in answering prompts based off a given clip. Such clips can be of desired length, bringing its versatility to both streaming and offline devices. However, due to the nature of videos, time must be taken into account as an important feature. Based on the time of the video, different frames may appear, which in turn may show different events. However, choosing the correct frames to look over in this scenario proves rather crucial. We define our problem statement in this area: *Can we use semantic understanding of a conditional question to select a subset of frames based off the query?*

Different models tend to solve this issue of selecting relevant frames in different ways. Some models use a Uniform Sampling approach, which takes evenly spaced frames, regardless of importance to the query. Scene-based techniques have been researched on, which group frames by scenes. The JCEF technique captions frames and uses an LLM to reason over such text Min et al. [2024]. We will dive deeper in such similar techniques in our *Related Work* section. In our paper, we will discuss a query-based approach that will select frames based off the question. To expand, we will focus on Conditional Questions, which are multi-step semantically.

2 Related Work

Frame selection based on a query has been researched in the past few years. Furthermore, there are other different techniques that can help group similar frames together. We model our technique based off on past frame selection methods. To further understand the background of frame selection, we must dive into the different techniques.

MoReVQA Min et al. [2024] implements a similar approach, as it creates a pipeline which consists of event parsing, visual grounding, LLM reasoning. It uses a frame captioning method, which is different to our pipeline. Apart from frame captioning, other techniques have used a frame clustering technique which can score clusters by query relevance, as seen in VideoTree Wang et al. [2024b]. Captioning techniques can be implemented into LLM reasoning techniques which use multi-round summarization prompts Zhang et al. [2024]. Zou et al. [2025] A.I.R, a VLM pipeline, uses an iterative loop which only applies VLM reasoning to a small subsection of candidate frames. Query-based selection of frames have been implemented in recent papers, where Submodular Mutual Information functions are utilized in choosing relevant and complementary frames. Patil et al. [2026]

Our pipeline takes methodological inspiration from such related work. In depth, it implements a query-based frame selection method. However, unlike Patil et al. [2026], it does not use Submodular Mutual Information function. Instead, it directly uses LLM reasoning that separates conditional phrases from a given Query. Thus, it clusters frames that answer the condition.

3 The CondVQA Dataset

The CondVQA Dataset includes 300 annotated videos, with annotations. Each annotation includes both a question, and an answer based on the question. The question will either be labeled as simple, or conditional. Multiple models will be tested on the same dataset, and the generated output will be compared with the annotated question/answer.

Furthermore, we will test CondVQA with a general benchmark. The MSVD benchmark Xu et al. [2017] (morpheusoc/msvd-qa) will be sampled based on 444 Video Question Pairs, pruning videos that have answers that either inaccurate, or questions that have multiple answers.

3.1 Data Collection

In order to create such dataset, we must sample videos that are of length 10-25 seconds. We will sample said videos from the PE Video Dataset Bolya et al. [2025]. The PE Video dataset is a large-scale collection of 1 million videos. The average video length is 17 seconds. We randomly collected 300 samples from the dataset, randomly downloading videos (in mp4 style) from the training/testing sets. Every mp4 file is then named to in 'tc[]' format, where [] signifies the number of the test case. We will then split the test cases into general and conditional.

3.2 Annotation Protocol

We annotate each sample video with an accurate analysis based on the video. If the video lack changes in scenes, then we classify the video as a potential candidate for a general question. A general question can be answered based on the whole video. For instance, asking for the colors visible in the video is a common general question utilized in the dataset. Since the question is classified as general, the Vision Language Model must look at all frames, and will answer the question based on inputted methodology (highest confidence, average confidence, etc).

Example of Conditional: When you see a basket, how many apples are inside the basket? (tc26)

Expected Answer: 4+ (based on frame)

If the video seems to show some sort of change in scenery or motion, we will likely classify the video as a potential candidate for a conditional question. We find a specific part of the video which may be unique, and ask a question about it. For example, if a man is wearing gloves, once he wears his gloves, the color of the gloves can be asked. This mentioned scenario can be classified as conditional, because only a subset of frames contain gloves. Thus, the algorithmic design is able to classify the subset of frames based on the condition of the question.

Example of General: What animal can be seen? (tc226)

Expected Answer: Tiger

Each video was manually annotated by the author, and checked for inconsistencies. There are cases that there may be multiple correct answers, or similar words will be accounted for. Questions that are not conditional or general, and that are past the scope of the project, are avoided. Each question is

a single sentence, with the expected answer to be no more than a handful of words. We will avoid using yes/no questions, as we want to focus more on descriptive answers.

3.3 Dataset Statistics

We split our data set into three partitions. The first split is a mix between conditional and general questions. The data set goes through tc1-tc200. There are 162 conditional questions and 38 general questions. This data set also includes an expected answer column. We will test CondVQA, Claude Opus 4.6, and Qwen2-VL-7B-Instruct Wang et al. [2024a] with this data set.

The second split is a dataset with 100 general questions. The data set goes through tc201-tc300. There are no conditional questions in this dataset. This data set also includes an expected answer column. We will test CondVQA, Claude Opus 4.6, and Qwen2-VL-7B-Instruct with this data set.

The third split is a data set with 444 sampled videos from a MSVD-qa repository in hugging-face. We will use this data set as a benchmark to compare our model in other cases. All questions in this dataset are classified and GENERAL, and are fairly short. We have pruned answers that are too general and incorrect. We will test CondVQA with this data set. (morpheushoc/msvd-qa).

Table 1: Dataset split statistics for CondVQA, Claude Opus 4.6, and Qwen2-VL-7B.

Split	Range	Conditional	General	Total
Split 1	tc1–tc200	162	38	200
Split 2	tc201–tc300	0	100	100
Total		162	582	744

All videos are less than 30 seconds, but more than 5 seconds. No video is accounted for sound, as we are only focusing on visual applications.

4 Model Architecture

We propose a hybrid LLM and Vision Language model. Our LLM focuses on parsing the question and classifying the question type. Given the question type is conditional, the LLM splits the full input into CONDITION and QUESTION. Subset selection is based on CONDITION, and answering the initial input is based on the QUESTION. We will use Qwen/Qwen2.5-1.5B-Instruct for Text Generation/Thinking, and llava-hf/llava-onevision-qwen2-0.5b-ov-hf Li et al. [2024] for image answering.

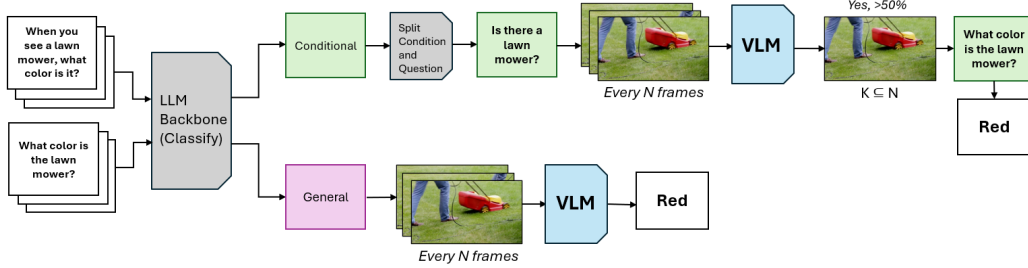


Figure 1: CondVQA pipeline for conditional and general visual question answering. We define N as the set of all frames in a given video input and $K \subseteq N$ as the conditional subset of frames, where $|N|, |K| \in \mathbb{Z}^+$.

4.1 Classifying Inputs

Our model takes in two inputs: a question and a video file. In sequential order, the model focuses on the question first. Suppose that we have a text input K , where K is a question based on the video

input. We classify such input K as a condition based on few-shot examples fed to the LLM backbone. We give the LLM an instruction and give it context based on its role in the model architecture.

Table 2: Question classification rules for CondVQA.

Type	Pattern	Output
CONDITIONAL	“when X, Y”, “where X, Y”, “when you see X, Y”	Condition (yes/no) + Question
SIMPLE	No condition, no timestamp	Question only

Examples:

- “When there is a dog in the frame, what color is the dog?” → CONDITIONAL
Condition: Is there a dog in the frame?
Question: What color is the dog?
- “When you see a number one, what color is it?” → CONDITIONAL
Condition: Is there a number one?
Question: What color is it?

The model’s ability to classify CONDITIONAL and GENERAL questions is based on the few-shot examples. Generally, keywords such as "when" help indicate conditions. When the question is classified as CONDITIONAL, the model expects there to be a condition input, and a basic question input. When the question is classified as SIMPLE, the model expects one input: a basic question.

4.2 Frame Subsets

Based on the previously explained input passed through, the model behaves differently depending on the classification. If the question input is classified as "CONDITIONAL", the conditional section of the question is extracted. That conditional section is then passed through the selection of frame subsets. The condition is subsequently rephrased into a simple question. For example, given "When there is a dog in the frame, what color is the dog?", the conditional section is "When there is a dog in the frame". Such phrase is then reformulated as: "Is there a dog in the frame?".

Once this question is passed, the model then examines every frame. It must be noted that this question, in specific, will only have two possible answers: yes or no. The Vision Language Model, with the input of the question, will output either yes or no based on the frame. For all N frames, it selects a subset K of frames, which all have high attribution scores for "yes". Now, we will pass the other portion of the original question. For example, given "When there is a dog in the frame, what color is the dog?", the remaining section is "What color is the dog?". As the frames passed should have a dog, the model now answers based strictly on relevant frames.

Otherwise, if the question input is classified as "SIMPLE", it will supervise all N frames. We then process each frame with the Vision Language Model, which generates an answer based on the specified parameters.

4.3 Vision Model

The vision model must be a Vision Language Model, meaning that it can answer questions given an image. Consequently, we do not need any model that processes video, as our model expands image processing to video understanding. It must be noted that, based on the performance of the vision language model, the performance of CondVQA may vary. In testing, we used llava-hf/llava-onevision-qwen2-0.5b-ov-hf. To test for sanity, we implemented integrated gradients in the Vision Language Model. Integrated Gradients Sundararajan et al. [2017] attributes the prediction of a deep network to its input features, measuring how each feature contributes to the model’s output.

4.4 Parameters

To run CondVQA, we must pass 12 parameters which will dictate the model’s behavior. The parameters can be slightly different, depending on whether one wishes to test the model with multiple

video-question pairs at a time. Our full parameter list can be found in our github, and the relevant use-cases are implemented in our google colab.

- `question`: String input of a question
- `video_dir`: Directory of the video
- `model`: Model of VLM
- `processor`: Type of CPU / GPU Processor
- `text_model`: Model of LLM
- `text_tokenizer`: Tokenizer of LLM
- `fps_sample`: Frames sampled per second
- `confidence_threshold`: Threshold for selecting the subset of frames
- `aggregation_method`: 'most_confident', 'consensus', or 'average'
- `show_visualizations`: Shows model's Integrated Gradient (regions which most influenced the VLM's answer)
- `save_results`: Saves results to a CSV file
- `output_dir`: Directory of saved result

5 Experimental Analysis

5.1 Experimental Setup

To test the accuracy of CondVQA, we will test it with our dataset mentioned in 3. Our dataset includes a mixture of general and conditional questions, showing the versatility of CondVQA. Thus, because of the size of our dataset, we must efficiently test the dataset in batches. To run the process, we will use an A100 GPU. Our model is able to read the question input, process each video, and output an answer in 30 seconds. Therefore, our 300 samples should take 3 hours of computation. For repeatability, we ran the code with Google Colab, meaning that with respective batch has its own cell. Given the mentioned parameters, a CSV file will be saved in said directory, with a `video_id` column, a question column, and an answer column.

We are testing CondVQA against Claude Opus 4.6, and Qwen2-VL-7B-Instruct. We will ask Claude, using Claude's chat bot, to answer videos based on a list of questions given. These questions are the same as the ones tested in CondVQA. We will be using Claude's Opus 4.6, which is Anthropic's strongest model to date, which has a focus of agentic coding, deep reasoning, and long-horizon task execution. As for Qwen2-VL-7B-Instruct, we will use a similar script to CondVQA's batch testing. Qwen2-VL-7B-Instruct is a video-answering model, meaning that we do not have to worry about any frame-splitting methodologies.

5.2 Evaluation Metrics

We will be calculating the accuracy of each model by finding the percent of correct answers over all answers. As mentioned, each question and answer is annotated by human, meaning that each video is carefully watched, and a question is created based on the content of the video. Afterwards, we manually answer the question. There are cases that some questions may have multiple answers. For instance, the question *"when you see birds, how many birds do you see?"* may have different answers depending on what frame the highest attribution comes from. Any answer that is in correct in at least one of the frames will be counted as a correct answer. Furthermore, answers that are very similar, such as *Teacher -> Instructor* will also be counted as correct. Incorrect answers will be answers that either do not answer the question, or answer the question incorrectly. Although very rare, if some model puts out an error for the question, then it will be counted as incorrect.

Furthermore, we will test CondVQA in a benchmark dataset. Specifically, we will use morpheushoc/msvd-qa. There will be 444 rows of question/video pairs. The dataset's answers are rather ambiguous, and in some cases very general. Thus, we will only count an answer incorrectly if there is no significance in the answer based on any of the frames. It must be noted that some of the original dataset was cut off due to the lack of clarity in some of the videos.

5.3 Results

The results, as mentioned, will be represented as a comparison for the first 300 samples. It must be reminded that the video test cases tc1-tc200 are a mixture of conditional and general, but the large majority (over 70%) is conditional. The video test cases tc201-tc300 are fully general.

Table 3: Accuracy comparison across models and dataset splits.

Model	tc1–tc200	tc201–tc300
CondVQA	181/200 (90.5%)	95/100 (95.0%)
Claude Opus 4.6	191/200 (95.5%)	97/100 (97.0%)
Qwen2-VL-7B	188/200 (94.0%)	97/100 (97.0%)

As visualized in the table, CondVQA competes with Claude and Qwen in accuracy. Although slightly less accurate, it is still above the 90th percentile. All three models perform better in the general question dataset, when compared to the conditional question dataset. CondVQA’s performance varies based on the Vision Language Model given. For instance, the Vision Language Model used in testing has around 500 million parameters. There are some Vision Language Models which have billions, meaning that they may perform better. As for the time taken to process all videos, Qwen2-VL-7B excels, going through all videos in less than half the time of the other two models. To further test the versatility of CondVQA, we test CondVQA with 3 benchmark datasets which contain only general questions. Claude and Qwen need not to be tested in such datasets, as the dataset has been widely used as a benchmark.

Table 4: Accuracy comparison across 3 benchmark datasets.

Benchmark Dataset	CondVQA
MSVD-QA Xu et al. [2017]	421/444 (90.5%)
MSRVTT Xu et al. [2016]	191/200 (95.5%)
YouCook2 Zhou et al. [2018]	198/200 (99.0%)

CondVQA scored a 421/444 in correct answers for the MSVD-QA, meaning that it has a very similar accuracy score to the general dataset. As MSVD-QA includes general questions, such correlation seems to be trivial. We then test our dataset of MSRVTT Xu et al. [2016], which CondVQA seems to answer very well compared to the expected annotated answers. Out of the 200 randomly sampled test cases, the answer CondVQA generates is similar 191 times, scoring a 95.5% accuracy. Finally, for our third benchmark (YouCook2 Zhou et al. [2018]), we asked it the same question for each sample, *"If you see food, what food do you see?"*. Thus, we were able to test for conditional accuracy in a benchmark dataset. We manually annotated each of the 200 randomly sampled videos, and CondVQA was correctly able to point out at least one food shown in the video 99% of the time.

5.4 Ablation Study

To further understand the advantageous of selecting a few given frames, we decide to conduct an ablation study. In specific, we have two different classes. The first class will be the same results from the original test, without any ablation. The second class will automatically classify every question as general, in other words ablating the Condition Aspect of the model. We create a new method, which forces every question to look as general. In depth, the basic frame splitting method that many VLM are capable of doing are similar to the general question archetype, where every frame is passed to the full question.

Table 5: Accuracy comparison across the full and ablated model.

Ablation Method	CondVQA
Complete	181/200 (90.5%)
Without Condition Extraction	127/200 (63.5%)

The results suggest that the full model performs better than the ablated model, with a considerable margin. Compared to the 90.5% accuracy rate, the ablated model was accurate only 63.5% of the time. To demonstrate statistical significance between the two proportions, we ran a two-proportion z -test. We define our null hypothesis $H_0 : p_1 = p_2$ and our alternative hypothesis $H_1 : p_1 \neq p_2$. The test yielded $z = -6.416$ and $p < 0.0001$, with a 95% confidence interval for the difference of $[-34.8\%, -19.2\%]$. We therefore reject H_0 and conclude that the full model achieves a statistically significantly higher accuracy than the ablated model ($p < 0.0001$).

6 Discussions

The results signify that CondVQA is on par with other models when it comes to both general and conditional questions. However, CondVQA’s methodology for answering such questions is unique. Most Vision Language Models, including Qwen2-VL-7B, samples videos per frames, and are processed as flat frame sequences with positional encoding. In other words, every frame is used in answering the full question.

CondVQA’s advantage lies in its noise reduction and its scaling in video length. It is clear that CondVQA only focuses on a certain subset of frames in a given video. Thus, it can be inferred that noisy frames (frames that are not important in answering the question) are not accounted for. In other models, every frame, regardless of whether it proves as useful, must be accounted for in the final output. In some cases, this proves helpful for our model.

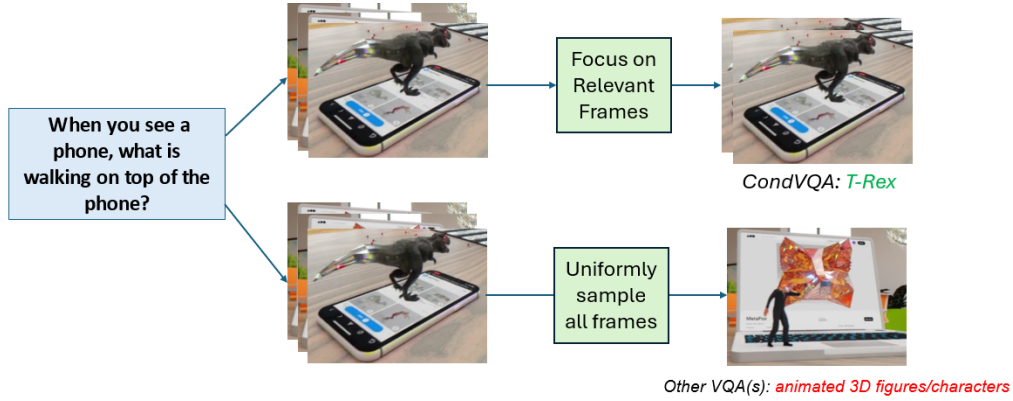


Figure 2: Example of how CondVQA’s frame extraction method may help reduce noise reduction. As it only focuses on the frames with a phone, it generates an accurate answer. We use tc47 for this example.

Although CondVQA has its benefits, much of its error can be attributed to its nature of looking at singular frames. In other words, if there are multiple right answers, it only looks at a certain frame, given a parameter. We can improve such constraint by further comparing each frame in the subset, showing possible changes throughout the frames. In the ablated model, it only looked at the full subset of singular frames. It was unable to answer many conditional questions due to the nature of the questions, and in many cases outputted answers that did not contain the condition. However, the ablated model does answer well in general questions, as seen in the benchmark tests, where every question in all benchmarks should be classified as "general".

CondVQA’s accuracy is also dependent on the given parameters: *text_model* and *model*. As mentioned above, *text_model* is the LLM backbone model, while *model* is the VLM model. We must acknowledge that, depending on the performance of the VLM, the responses of CondVQA may vary. For our experimental setup, we implemented llava-hf/llava-onevision-qwen2-0.5b-ov-hf Li et al. [2024] for image answering. It must be noted that this model has 0.5 billion parameters, which can be replaced for a more sophisticated model.

Throughout our testing process, some of our limitations in terms of running time is attributed to our decision to output the model’s thinking process. Mentioned processes include splitting the conditional

section off the question, along with subset selection of mentioned frames. We can also attribute the running time based on the Vision Language Model that was used.

7 Conclusion

We conclude that CondVQA can work on-par with other models, when facing conditional questions. The naive solution that CondVQA offers seems to be effective, and ablation studies further prove such claim. Using pure semantic understanding of conditional questions, we can proceed to frame selection. CondVQA’s pipeline distinguishes queries, and is able to classify a question as conditional or temporal. Thus, we were able to test CondVQA in both conditional and general datasets, which contain conditional, and general questions, respectively. By LLM reasoning, it not only classifies questions, but classifies conditions. If a condition is found, it then clusters frames that are true based off the condition.

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